

UTOPIA PLANITIA

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TYPE Lowland plain
AGE 2–3.5 billion years
DIAMETER 2,000 miles
 (3,200 km)

Utopia is one of the enormous lava-covered plains of the northern hemisphere. The giant Elysium volcanoes are at its eastern perimeter. From above, it is possible to see that complex albedo patterns, polygonal fractures, and craters mark the vast rolling plain. Down on the surface, the landscape is uniformly flat and rock-strewn. At least that is the view in northeastern Utopia, where Viking 2 landed on September 3, 1976. Angular boulders of basaltic rock cover the landing site, close to Mie Crater. Small holes in the rocks are

a result of bursting bubbles of volcanic gas. A thin layer of frost was also seen by the craft, first in mid-1977, when it covered the surface for about 100 Earth days, and then when it built up again in May 1979, 1 Martian year (23 Earth months) later.

These giant polygons are not unique to Utopia; they are also seen on other northern plains such as Acidalia and Elysium (below). They are polygon-shaped chunks of flat land separated by huge cracks, or troughs, and are reminiscent of mud cracks seen in dried-up ponds on Earth. Earth's polygonal patterns are book- to table-sized; the Martian



POLYGON TROUGH FLOOR

This close-up of one of the huge surface cracks that isolate polygon-shaped areas of land reveals bright, evenly spaced windblown ripples of sediment.

ones are on a far grander scale, the size of a town or small city. The land areas are 3–12 miles (5–20 km) across, and the cracks between them are hundreds of yards wide. Earth's mud cracks form as ground dries through water evaporation. The Martian cracks could have a similar origin. Certainly, Mars

has experienced the large-scale floods of water required. And dust-covered cemented rock found by Viking 2 seemed to be held together by salts left behind as briny water vaporized. However, it has also been suggested that the polygons formed in other ways—for example, in cooling lava.

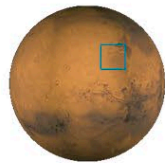


ICE ON THE ROCKS

A coating of water ice covers the volcanic rocks and soil at the Viking 2 landing site. The ice layer is incredibly thin, no more than a thousandth of an inch (a fraction of a millimeter) thick.

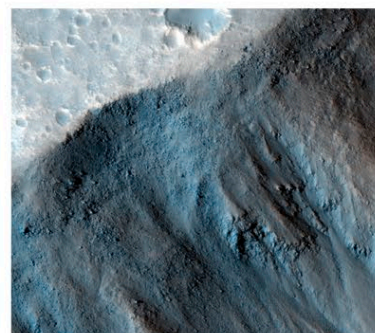
LUNAE PLANUM

Kasei Valles



TYPE Outflow channel
AGE 3–3.5 billion years
LENGTH 1,105 miles
 (1,780 km)

Kasei, which takes its name from the Japanese word for Mars, is the largest outflow channel. Not only is it long, but parts of its upper reaches are over 125 miles (200 km) across, and in places, it is over 2 miles (3 km) deep. The catastrophic flooding that formed Kasei was greater than any other known flood event on Mars or Earth. Kasei originates in Lunae Planum, directly north of central Valles Marineris (see p.158), then flows across the ridged plain to Chryse Planitia. Along its route lie streamlined islands, isolated as the water flow split and then rejoined.



PLATEAU EDGE

This image shows the steep edge of a valley in northern Kasei Valles. The plateau to the left is about 0.8 miles (1.3 km) higher than the floor of the valley, similar to the depth of the Grand Canyon on Earth.

ELYSIUM PLANITIA

Elysium Planitia



TYPE Lowland plain
AGE Under 2.5 billion years
DIAMETER 1,860 miles
 (3,000 km)

The Elysium Planitia is an extensive lava-covered plain just north of the equator. It has been suggested that an area almost directly south of the great volcano Elysium Mons is a dust-covered frozen sea. It is dominated by irregular blocky shapes that look like the rafts of segmented sea ice seen off

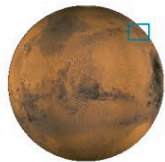
the coast of Earth's Antarctica. These "ice plateaus" are surrounded by bare rock. They formed when water flooded through a series of fractures in the Martian crust, creating a sea similar in size to Earth's North Sea. As the water froze, floating pack ice broke into rafts. These were later covered in dust from the nearby volcanoes, and this coating protected them. Unprotected ice between the rafts vaporized into the atmosphere, leaving bare rock around the ice plateaus.

ICE PLATEAUS AND IMPACT CRATERS

The darker-toned ice plateaus are a few tens of miles across. The relatively small number of impact craters in this area suggests a young surface.

PROTONILUS MENSÆ

Rock-strewn Glacier



TYPE Valley glacier
AGE Under 1 billion years
LENGTH 5 miles
 (8 km)

Many glaciers have been mapped across the midlatitudes of Mars. The region known as Protonilus Mensae, which is about 40° north of Mars's equator, hosts some of the nicest examples. They are not as obvious as you might expect, because the bright white ice is hidden below rock debris and dust. However, the flow lineations on their surfaces and the built-up debris (moraine) in front make them convincing evidence of an epoch when accumulated snowfall on Mars allowed glaciers to form and carve out valleys.

GLACIAL EROSION

This glacier, flowing out from a region of high ground, has widened and deepened what was probably originally a much narrower valley.

